

Lead DS Experimentation Playbook — From Essentials to Advanced

A practical, math-friendly guide you can hand to any data scientist—from first experiment to advanced causal inference—covering design, power, inference, monitoring, and what to do when an A/B test isn't possible.

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1) Mindset & Principles

- **Causality over correlation.** Experiments aim to estimate $\tau = E[Y(1) - Y(0)]$ for a clearly defined unit and time horizon.
 - **Pre-register your plan.** Fix metrics, windows, primary hypothesis, analysis code, and stopping rule before launch to avoid p-hacking.
 - **Simplicity first.** Start with the minimal design that answers the decision question; layer sophistication only when needed.
 - **Decisions, not p-values.** Tie your outcome to a decision rule (ship, iterate, stop) based on effect size, uncertainty, and guardrails.
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2) Causal Basics & Assumptions

2.1 Notation

- Potential outcomes: for unit i , $Y_i(1)$ if treated, $Y_i(0)$ if control. Observed $Y_i = D_i Y_i(1) + (1 - D_i) Y_i(0)$ with $D_i \in \{0, 1\}$.
- Average Treatment Effect (ATE): $\tau = E[Y(1) - Y(0)]$.

2.2 SUTVA (Stable Unit Treatment Value Assumption)

- **No interference:** one unit's treatment doesn't affect another's outcome.
- **No hidden versions of treatment:** treatment is well-defined.

When SUTVA breaks: network effects, marketplace spillovers, shared caches, messaging/virality, geo cannibalization.

Mitigations: cluster randomization (geo, network clusters), switchback by time, two-stage designs (randomize encouragement then exposure), interference-aware estimators, holdout geos.

2.3 Ignorability / Unconfoundedness

- Randomization ensures $D \perp (Y(0), Y(1))$, making difference-in-means unbiased.

2.4 Consistency & Positivity

- **Consistency:** observed outcome equals the potential outcome under received treatment.
- **Positivity (overlap):** every unit has nonzero probability for each arm.

3) Choosing the Experimental Unit & Bucketing

- **Unit:** user, session, device, cookie, merchant, courier, search, impression, request, or geo.
- **Rule of thumb:** randomize at the **highest level where SUTVA plausibly holds** and assignment can be enforced.
- **Bucketing:** deterministic hashing (e.g., $\text{murmurhash}(\text{user_id}) \bmod 100$) to create stable, mutually exclusive buckets; define **exposure** precisely to avoid partial-treatment bias.
- **Collision risks:** cross-device identity, shared accounts, or cross-product features -> consider account-level assignment or identity resolution.

4) Designs

4.1 A/A

- Purpose: check randomization, metric stability, variance estimates, analytics pipeline. Expect 5% false positives at $\alpha = 0.05$.

4.2 A/B(/n)

- Parallel arms with fixed allocation (e.g., 50/50). Scales to A/B/n but watch **multiple testing**.

4.3 Factorial & MVT

- Test multiple factors simultaneously. For 2×2, estimate main effects and interaction:

$$Y = \beta_0 + \beta_A A + \beta_B B + \beta_{AB} AB + \epsilon.$$

Pros: efficient; **Cons:** interpretation & power for interactions.

4.4 Switchback (Time-based)

- Alternate treatment/control by time blocks (e.g., hours or days); good for platform-wide changes with interference (e.g., pricing, ranking). Use blocks larger than autocorrelation scale. Analyze with paired or cluster-robust SEs.

4.5 Cluster / Geo Randomization

- Randomize geos/stores/regions to contain spillovers. Power depends on **number of clusters** and **ICC (intra-cluster correlation)**.

4.6 Crossover

- Units receive sequences (e.g., ABBA). Requires no carryover; add washout periods.

4.7 Staggered Rollout / Stepped Wedge

- Gradually treat clusters; analyze with difference-in-differences (DiD) and cluster-robust SEs.

5) Metrics & OEC

- **North Star / OEC:** a single primary metric used for ship/no-ship.
- **Guardrails:** safety metrics (e.g., crash rate, delivery time, fraud) with thresholds.
- **Leading vs Lagging:** choose a window that aligns with decision latency.

5.1 Metric Construction

- **Binary** (conversion), **rate** (clicks/impressions), **count** (orders), **value** (revenue), **ratio** (ARPU), **quantiles** (p95 latency).
- Use **cohorting** to avoid double counting.
- For ratios use **Delta method** or **Fieller**; consider **log ratio** transformation.

5.2 Variance Behavior

- Heavy tails (revenue) → trimming/winsorization, log-transform, robust estimators, or quantile-based outcomes.
- Serial correlation in time → cluster-robust SEs.

6) Sample Size & Power

Let power $1 - \beta$, type I error α , Minimal Detectable Effect (MDE) δ . Two-arm parallel design; allocation p to treatment.

6.1 Continuous Outcome (difference in means)

Assume common variance σ^2 , total N units.

$$N = \frac{(z_{1-\alpha/2} + z_{1-\beta})^2 2\sigma^2}{\delta^2} \quad \text{for 50/50 split}$$

General with allocation p :

$$N = \frac{(z_{1-\alpha/2} + z_{1-\beta})^2 \sigma^2}{\delta^2 p(1-p)}.$$

Estimate σ from historical data or an A/A. Adjust upward for clustering (design effect) and SRM risk.

6.2 Binary Outcome (difference in proportions)

Baseline rate p_0 , target uplift Δ . Pooled variance approx:

$$N = \frac{(z_{1-\alpha/2} + z_{1-\beta})^2 [p_0(1-p_0) + p_1(1-p_1)]}{\Delta^2} \quad \text{with } p_1 = p_0 + \Delta.$$

6.3 Ratios & Logs

For ratio $R = \bar{Y}/\bar{X}$, use **Delta method**:

$$\text{Var}(R) \approx \frac{1}{\mu_X^2} \text{Var}(\bar{Y}) + \frac{\mu_Y^2}{\mu_X^4} \text{Var}(\bar{X}) - 2 \frac{\mu_Y}{\mu_X^3} \text{Cov}(\bar{X}, \bar{Y}).$$

Often easier to work with log ratios.

6.4 Clustering & ICC

Design effect: $DE = 1 + (m - 1)\rho$ where m is average cluster size, ρ ICC. Effective sample size $N_{\text{eff}} = N/DE$. Inflate sample size by DE .

6.5 Power for Percent Change

If target is δ relative: define on log scale: $\delta_{\log} = \log(1 + \delta)$ and use continuous formula with log-transformed outcome.

7) Randomization, Stratification, CUPED & Variance Reduction

7.1 Plain Randomization

Use deterministic hashing on unit IDs to avoid collisions, with stable exposure mapping.

7.2 Stratified / Block Randomization

Stratify by high-variance covariates (geo, platform, activity tier). Ensures balance and **power gains**. Analyze with stratification-aware estimators (include stratum fixed effects).

7.3 Re-randomization (Rerand)

Repeat randomization until covariate balance criteria met (Mahalanobis distance threshold). Keep analysis accounting for the rerandomization scheme.

7.4 CUPED (Covariate-Adjusted)

Use a pre-experiment covariate X correlated with outcome Y . Define

$$\tilde{Y} = Y - \theta(X - \bar{X}) \text{ with } \theta = \frac{\text{Cov}(X, Y)}{\text{Var}(X)}.$$

Run the standard A/B on $\tilde{Y}$. **Variance reduction** $\approx$ correlation² of X with Y .

7.5 Other Adjustments

- Linear regression with pre-periods and covariates; cluster-robust SEs.
- Pairing/matching (e.g., for geos) then paired t-tests.
- Post-stratification weighting if arms imbalanced on known covariates.

8) Estimation: Effects, Uncertainty & Ratio Metrics

8.1 Difference in Means (User-level)

$\hat{\tau} = \bar{Y}_T - \bar{Y}_C$. Use robust (HC) or cluster-robust SEs if needed.

8.2 Percent Change

$\% \Delta = \frac{\bar{Y}_T - \bar{Y}_C}{\bar{Y}_C}$. For inference, prefer log transform: estimate $\Delta_{\log} = E[\log Y|T] - E[\log Y|C]$, then back-transform.

8.3 Ratio Metrics

- E.g., ARPU = revenue/users. Use Delta method, Fieller's theorem, or bootstrap. If denominator can be zero, avoid naive ratios; prefer per-user aggregates then across users.

8.4 Robustness to Heavy Tails

- **Winsorize/trim** top x%; report sensitivity.
- **Huber M-estimators** for robust means.
- **Quantile treatment effects** via quantile regression.

8.5 Interval Estimation

- Prefer **confidence intervals** with explicit level (e.g., 95%). For binary/rare events, use **Wilson** or **Agresti–Coull** intervals; for ratios consider Fieller/bootstraps.
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9) Hypothesis Tests: Catalog & Guidance

9.1 Parametric

- **Two-sample t-test (pooled)**: equal variances;
- **Welch t-test**: unequal variances (default for means).
- **Z-test**: large samples with known variance or via CLT on means.
- **Chi-square test**: independence or goodness-of-fit (counts);
- **Fisher's exact**: small counts.

9.2 Nonparametric / Randomization-based

- **Mann–Whitney (Wilcoxon rank-sum)**: shift in distributions.
- **Permutation test**: exact under exchangeability; gold standard with arbitrary metrics if compute budget allows.
- **Sign/Wilcoxon signed-rank**: paired designs.

9.3 Bootstrap (Resampling)

- **Percentile CI**: simple but biased under skew.
- **BCa** (bias-corrected & accelerated): better coverage.
- **Studentized**: bootstrap t.
- Use **stratified bootstrap** for cluster/switchback; **block bootstrap** for time-series.

9.4 Choosing a Test

- Means with unequal variances → **Welch**.
 - Proportions → **two-proportion z** or **Fisher** if small.
 - Arbitrary/ratio/complex metric → **permutation** or **bootstrap** + Delta method as check.
 - Clustered/time → cluster-robust or block bootstrap.
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10) Sequential Analysis & Early Decisions

10.1 The Peeking Problem

Repeated looks inflate type I error. **Do not** repeatedly peek with fixed-horizon tests.

10.2 Alpha-Spending (Group-Sequential)

Define interim looks with spending function $\alpha(t)$. Common: **O'Brien-Fleming** (conservative early), **Pocock** (equal). Guarantees FWER control.

10.3 Always-Valid Inference

Use tests that remain valid under continuous monitoring: **mSPRT**, **e-values/nonnegative supermartingales**, **betting/sequential probability ratio**. Report **always-valid p** or **confidence sequences** CI_t that hold uniformly over time.

10.4 Robust “Sign” Detection (Early Pick)

- Use **confidence sequences** for the effect τ . If the lower bound of the 95% CS > 0 (or upper < 0), you can declare sign **at that time** with Type I controlled.
- Alternatively, **Bayesian decision** with posterior probability of benefit (e.g., $P(\tau > 0 | \text{data}) > 0.975$) and **expected loss** threshold.
- For directional decisions without shipping, use **non-inferiority** or **minimum effect** tests with sequential control.

11) Multiple Testing

11.1 Family-Wise Error Rate (FWER)

- **Bonferroni**: test at α/m . Simple, conservative.
- **Holm (step-down)**: stronger than Bonferroni, controls FWER.
- **Hochberg (step-up)**: more powerful (requires independence/PRDS).

11.2 False Discovery Rate (FDR)

- **Benjamini-Hochberg (BH)**: controls FDR under independence/PRDS.
- **Benjamini-Yekutieli (BY)**: conservative under arbitrary dependence.

11.3 Structured Corrections

- **Gatekeeping/hierarchical testing**: test primary first; only proceed if significant.
- **Westfall-Young permutation** for strong control with complex dependence.
- **Selective inference** adjustments if hypotheses chosen adaptively.

Define the family explicitly (e.g., all variants within an experiment, or all metrics tested in a decision).

12) Launch Monitoring & Validity Checks

12.1 SRM (Sample Ratio Mismatch)

Check allocation vs expected using chi-square on counts. Flag if $p < 0.001$; investigate logging/exposure bugs.

12.2 Attrition & Compliance

- **Attrition**: different missingness across arms → bias.
- **Noncompliance**: ITT vs TOT; instrument with assignment to estimate **LATE** if necessary.

12.3 Instrumentation & Data Quality

- Event schema changes, delayed logging, duplicate events, bots/fraud.
- Metric drift vs definition drift. Maintain a **metric contract**.

12.4 Time & Seasonality

- Day-of-week/season effects: ensure balanced windows, or include time fixed effects / block designs.
- Novelty/learning effects: require minimum exposure duration.

12.5 Guardrails in Monitoring

- Crash/error rates, latency tail (p95), cancellations, chargebacks, partner churn; stop rules if guardrails breached.

13) When Analysis Is Not Valid (A/B Not Reliable)

- **SRM detected** → stop and fix randomization/exposure.
- **Interference** → re-design with cluster/switchback; don't analyze as iid.
- **Identity leakage/collisions** → re-key unit or deduplicate.
- **Time-varying rollout with naive analysis** → use DiD/segmented regression.
- **Severe heavy tails** → robust estimation or transform; avoid mean-based tests alone.
- **Peeking with fixed-horizon** → re-analyze with sequential methods or hold-out final window and pre-specify.

14) When A/B Isn't Possible

14.1 Quasi-Experiments

- **Difference-in-Differences (DiD)** with treated/untreated geos and time FE; check **parallel trends**.
- **Interrupted Time Series** / segmented regression for platform-wide changes.
- **Synthetic Control** for one/few treated geos.

14.2 Observational Causal Inference

- **Propensity score** matching/weighting;
- **Outcome regression**;
- **Doubly Robust (AIPW)**: combines both, consistent if either model correct.
- **Instrumental Variables (IV)** when you have exogenous instrument.
- **Regression Discontinuity (RD)** near thresholds.

14.3 Marketplaces & Networks

- **Two-sided markets**: treat one side, measure on both; cluster by market; beware spillovers and equilibrium shifts.

15) Bandits, Offline Policy Evaluation, and Heterogeneous Effects

15.1 Bandits vs A/B

- **A/B**: best for estimation & learning; fixed allocation; clean inference.
- **Bandits**: optimize reward during test (e.g., Thompson sampling, UCB); harder inference, use posterior/empirical Bayes or always-valid CS for monitoring.

15.2 Offline Policy Evaluation (OPE)

For logged bandit/recommendation data:

- **IPS**: $\hat{V}_{IPS} = \frac{1}{N} \sum \frac{\pi(a|x)}{\mu(a|x)} r$; high variance.
- **Self-normalized IPS** and **Doubly Robust (DR/AIPW)** reduce variance.
- Requires known logging policy μ and support (positivity).

15.3 Uplift & Heterogeneous Treatment Effects (HTE)

- Guard against p-hacking in subgroup hunts; pre-specify.
- Methods: **causal trees/forests**, **uplift modeling**, **T-Learner/S-Learner/X-Learner**.
- Validate with **honest splitting** and **policy value** evaluation.

16) Practical Checklists & Templates

16.1 Pre-Experiment Checklist

- [] Decision question & success criteria (ship/no-ship).
- [] Primary metric (OEC) + guardrails; definitions frozen.
- [] Unit of analysis; exposure and eligibility.
- [] Design & allocation; blocking/stratification.
- [] Sample size & power; MDE.
- [] Variance reduction plan (CUPED/covariates).
- [] Analysis plan (estimators, intervals, tests).

- [] Stopping rule (fixed horizon / group sequential / always-valid).
- [] Multiple testing plan (Holm/BH, gatekeeping).
- [] Data quality & SRM monitoring.
- [] Guardrail stop thresholds.
- [] Pre-register plan & freeze code.

16.2 During-Experiment Monitoring

- [] SRM ($p < 0.001$ triggers investigation).
- [] Compliance/exposure distributions stable.
- [] Guardrails within bounds.
- [] Sequential thresholds respected.

16.3 Post-Experiment

- [] Primary effect + 95% CI + decision.
- [] Sensitivity: robust/winsorized, CUPED, bootstrap, permutation.
- [] Subgroup analyses (pre-registered) + multiplicity control.
- [] Learnings, follow-ups, and registry entry.

17) Appendix: Math, Code & Reference

17.1 SRM Test (Chi-square)

Given expected allocation π_T, π_C and observed counts n_T, n_C :

$$\chi^2 = \sum_{g \in \{T, C\}} \frac{(n_g - n\pi_g)^2}{n\pi_g} \text{ with 1 d.f.; flag if } p < 0.001.$$

Python:

```
import numpy as np
from scipy.stats import chi2
n_T, n_C = 505_100, 494_900
pi_T, pi_C = 0.5, 0.5
n = n_T + n_C
chi2_stat = ((n_T - n*pi_T)**2)/(n*pi_T) + ((n_C - n*pi_C)**2)/(n*pi_C)
pval = 1 - chi2.cdf(chi2_stat, df=1)
```

17.2 Welch Test for Means

Test statistic:

$$t = \frac{\bar{Y}_T - \bar{Y}_C}{\sqrt{s_T^2/n_T + s_C^2/n_C}},$$

$$\nu \approx \frac{(s_T^2/n_T + s_C^2/n_C)^2}{(s_T^2/n_T)^2/(n_T-1) + (s_C^2/n_C)^2/(n_C-1)}.$$

17.3 Two-Proportion z-Test

$$\hat{p}_g = x_g/n_g, \text{ pooled } \hat{p} = (x_T + x_C)/(n_T + n_C) .$$
$$z = \frac{\hat{p}_T - \hat{p}_C}{\sqrt{\hat{p}(1-\hat{p})(1/n_T + 1/n_C)}} .$$

17.4 CUPED Coefficient

$$\theta = \frac{\text{Cov}(X, Y)}{\text{Var}(X)} . \text{ Estimate via pre-period data; use cross-fitting to avoid overfit.}$$

17.5 Ratio via Fieller

For means $\bar{Y}, \bar{X}$ with covariance matrix Σ , CI for $R = \bar{Y}/\bar{X}$ solves a quadratic inequality (see Fieller's theorem). Prefer log-ratio or bootstrap in practice.

17.6 Delta Method for Log Percent Change

If $Z = \log \bar{Y}$,

$$\text{Var}(Z) \approx \frac{\text{Var}(\bar{Y})}{\mu_Y^2} .$$

Then CI on $\Delta_{\log}$ back-transforms to percent.

17.7 Block/Cluster-Robust SEs

For switchback/cluster: use regression $Y_{ig} = \alpha + \beta D_{ig} + \gamma_g + \epsilon_{ig}$ with cluster-robust SEs on g (block/time or geo).

17.8 Confidence Sequences (Always-Valid)

For mean μ with sub-Gaussian proxy σ , one can construct time-uniform CIs:

$CI_t = [\hat{\mu}_t \pm w(t, \alpha)]$, where width shrinks with t and validity holds for all stopping times. Practical implementations available in modern libraries.

17.9 Multiple Testing Recipes

- **Bonferroni**: reject if $p_i \leq \alpha/m$.
- **Holm**: order p's ascending; find smallest k with $p_{(k)} > \alpha/(m - k + 1)$; reject all up to $k - 1$.
- **BH (FDR)**: order p's; find largest k with $p_{(k)} \leq (k/m)\alpha$; reject all with rank $\leq k$.

17.10 Permutation Test Skeleton

```
import numpy as np

def perm_test(x, y, B=10000, stat=lambda a,b: np.mean(a)-np.mean(b),
             alternative='two-sided'):
    n, m = len(x), len(y)
    z = np.concatenate([x, y])
```

```

obs = stat(x, y)
cnt = 0
for _ in range(B):
    np.random.shuffle(z)
    x_p, y_p = z[:n], z[n:]
    s = stat(x_p, y_p)
    if alternative=='two-sided':
        cnt += abs(s) >= abs(obs)
    elif alternative=='greater':
        cnt += s >= obs
    else:
        cnt += s <= obs
p = (cnt+1)/(B+1)
return obs, p

```

17.11 Bootstrap CI (BCa Outline)

- Resample with replacement; compute statistic each time; adjust for bias & acceleration using jackknife. Prefer libraries for correctness.

17.12 Difference-in-Differences (2×2)

$$Y_{gt} = \alpha + \beta \text{Treated}_g + \gamma \text{Post}_t + \delta(\text{Treated} \times \text{Post}) + \epsilon_{gt}.$$

δ is the DiD estimate; use cluster-robust SEs by group.

17.13 Uplift Modeling Metric

Use **Qini** or **uplift at k%**; evaluate with a randomized holdout or via policy value with IPS on treatment propensities.

Teaching Narrative: From First Experiment to Advanced

- 1) Start with a clean A/B: define unit, OEC, guardrails, sample size; run an **A/A** to validate SRM and variance.
 - 2) Add **CUPED** with pre-period covariates; stratify by key axes (platform/geo).
 - 3) Use **Welch + bootstrap** for heavy-tailed metrics; report percent-change on log scale.
 - 4) If you must peek, adopt **group-sequential** or **always-valid** methods.
 - 5) Control multiplicity with **Holm** for families of endpoints or **BH** when screening many variants.
 - 6) For network/marketplace features, consider **cluster** or **switchback** designs.
 - 7) When randomized tests are infeasible, reach for **DiD/synthetic control** or **AIPW/IV/RD** with careful diagnostics.
 - 8) For ranking/policy changes, leverage **offline OPE** then carefully ramp with guardrails.
 - 9) Explore **HTE** only after main effect is decided; validate with honest splits and policy value.
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Common Pitfalls & Anti-Patterns

- Measuring at the wrong level (session instead of user) → inflated n, false precision.
 - Ignoring noncompliance/exposure (ITT ≠ TOT).
 - Using mean on extreme heavy tails without robustness.
 - Peeking with fixed-horizon p-values.
 - Undefined “family” for multiplicity → silent alpha inflation.
 - Treating SRM as a nuisance instead of a stop condition.
 - Not documenting metric definitions; silent schema changes.
 - Overfitting uplift models without honest evaluation.
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Minimal Templates

Experiment Spec (one-pager): - Objective & hypothesis (with effect direction & MDE).

- Unit & eligibility; randomization details; allocation.
- Metrics: primary, guardrails, secondary; definitions.
- Sample size & duration; assumptions.
- Analysis plan: estimator, CI, test; variance reduction; multiplicity; stopping rule.
- Monitoring: SRM rule, guardrail thresholds.
- Rollout/kill criteria; ramp plan.
- Owners; code & dashboards links.

Post-Readout: - Result summary (effect + CI); decision; counterfactual.

- Validity checks (SRM, compliance, DQ).
 - Sensitivity analyses; learnings; next steps.
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Final Notes

- This playbook is intentionally math-forward but pragmatic. Pair it with your internal training code (power calculators, CUPED builders, bootstrap/permutation utilities, SRM checks) and embed the templates into your experiment registry.